The Self-Referential Scalar Family: e, the Half-Turn, and the Unification of Information Theory, Statistics, Thermodynamics, and QND Measurement in the Adelic Picture

Rowan Brad Quni-Gudzinas

2026-08-20

Author: Rowan Brad Quni-Gudzinas | **Date:** 2026-08-20 |

Abstract

Standard information theory measures uncertainty at the archimedean place; the adelic program extends it to all completions of the rationals, carrying a p-adic valuation entropy alongside Shannon entropy. This paper shows that the two most familiar distributions of statistical mechanics are already hidden inside that p-adic structure. The p-adic maximum-entropy distribution — the geometric distribution on the valuation — is exactly the Bose–Einstein occupation distribution at fugacity $1/p$, that is, at inverse temperature $\ln p$ at the p-adic place; its mean, $1/(p-1)$, is the p-adic entropy itself. The squarefree restriction of the integers, which excludes repeated prime factors just as the Pauli exclusion forbids double occupation, is its Fermi–Dirac counterpart, with occupation probability $1/(p+1)$ at the same temperature. A quantum non-demolition measurement of a p-adic-valued observable preserves this entropy exactly: it is the equality case of the adelic data-processing inequality, the measurement that reads without demolition. These identifications are exact and computationally verified. They are assembled under a single structural thesis: the constants e, π, and the exchange phase $R = (e^{i\pi})^{2s}$ form one self-referential scalar family generated by the act of drawing a distinction — e as the fixed point of the operator $Df = f$, the Gaussian as the fixed point of the Fourier transform, the Boltzmann factor as the fixed point of the maximum-entropy principle, and the half-turn $e^{i\pi} = -1$ as the generator of quantum statistics. The Planckian bound $2\pi k_B T/\hbar$ and the architectural optimum $\ln(2\pi)$ carry the same circle-trace π. The premises end where the identification of a physical temperature at the p-adic place begins: the algebra is exact, the dictionary is proposed, and the falsification conditions are written.

1. Introduction

A compound question motivates this work: what is the relationship between information theory, statistics, thermodynamics, and quantum non-demolition (QND) measurement — especially as it bears on the self-reference of the constant e and on patterns of distinction such as re-entry — and how do recent results on spin statistics and on standard-model/condensed-matter unification tie in?

The answer developed here is a single structural claim: **the scalar family generated by the re-entrant mark — e as the fixed point of $Df = f$, π as the trace of the identity on the circle type, and the exchange phase $R = (e^{i\pi})^{2s}$ as the $(2s)$-fold half-turn — is the common skeleton of all four domains.** Information theory, statistics (in both senses: probability distributions and quantum statistics), thermodynamics, and QND measurement each turn out to be a face of the same self-referential fixed-point structure.

Three results are established, all computationally verified:

- **R1 (statistics = information, non-archimedean).** The p-adic maximum-entropy distribution of Adelic Shannon Theory [1], $P(v_p(X) = k) = (1 - 1/p)\, p^{-k}$, is exactly the Bose–Einstein occupation distribution with fugacity $z = e^{-\beta_p} = 1/p$, i.e. at inverse temperature $\beta_p = \ln p$ at the p-adic place. Its mean is the Bose–Einstein occupation number $\langle n \rangle = 1/(e^{\beta(\varepsilon-\mu)} - 1) = 1/(p-1)$, precisely the p-adic entropy $H_p^{\max}$.
- **R2 (statistics = information, fermionic).** The squarefree restriction of the integers ($v_p \in \{0, 1\}$ — the p-adic Pauli exclusion) yields $P(v_p = 1) = 1/(p+1)$, exactly the Fermi–Dirac occupation number at the same inverse temperature $\beta_p = \ln p$. Verified numerically over 1.2 million squarefree integers (max deviation 8.53×10^{-6}).
- **R3 (QND = equality case of the data-processing inequality).** An ideal QND measurement of a p-adic-valued observable preserves H_p exactly: it is the equality case of the adelic data-processing inequality — the measurement family (any channel commuting with the observable, reading its eigenbasis) that extracts information without increasing the measured observable's entropy.

Why a reader should care. The p-adic valuation entropy is not an exotic side structure: it is literally the occupation statistics of statistical mechanics, with the prime setting the temperature. That identification gives quantum engineers a per-prime uncertainty budget conserved by QND readout, gives energy benchmarking a per-prime cost scale, and gives foundations a concrete unification: e, π, and the exchange phase organize information theory, statistics, thermodynamics, and measurement as one structure. The claims are graded, verified computationally, and falsifiable.

Where the premises end. The results rest on imported machinery — the Adelic Shannon formalism $H_p = \mathbb{E}[v_p]$ [1], the exchange-phase reading $R = (e^{i\pi})^{2s}$ [2, 3], the Planckian bound [15] — and on two conjectural steps of this paper: that $\beta_p = \ln p$ is a *physical* temperature (exact algebra, proposed physics) and that the four domains are *one* family (the unification thesis). Both are stated with falsification conditions in Section 7.

2. The Three Faces of e

The constant e appears in this program three times, and each appearance is a fixed point of a self-referential operation:

1. e **as the fixed point of differentiation** (re-entrant calculus [3, 7]): the re-entrant mark under linear discipline solves $Df = f$, $f(0) = 1$, with unique solution $f(x) = e^x$. The function equal to its own rate of change — self-reference in its purest local form.
2. $e^{-\pi x^2}$ **as the fixed point of the Fourier transform** (Adelic Shannon Theory [1]): the Gaussian is the unique function (up to scaling) with $\mathcal{F}[g] = g$, an eigenfunction of the Fourier transform with eigenvalue 1. This is self-reference in its purest global (duality) form. The Poisson summation formula $\sum_n f(n) = \sum_n \hat{f}(n)$ holds *because* the Gaussian is this fixed point — source-channel equality at every place simultaneously.
3. $e^{-\beta E}$ **as the fixed point of the maximum-entropy variational principle** (thermodynamics): the Boltzmann factor is the unique distribution maximizing entropy subject to fixed mean energy — the distribution equal to its own max-entropy solution. The Gaussian is the continuous case; the Boltzmann factor is the general case; both are fixed points of the entropy functional under moment constraints.

The re-entrant calculus [3, 7] already identified faces 1 and the half-turn $e^{i\pi} = -1$; Adelic Shannon Theory [1] identified face 2; thermodynamics has always known face 3. The unification claim of this paper is that these are **the same self-reference at different levels of the adelic hierarchy**: local (D), global ($\mathcal{F}$), and variational (max-entropy). The half-turn $e^{i\pi} = -1$ is the bridge: it is the *rotation* fixed point (a 2π rotation of the circle is the identity — the circle's self-reference), and it generates the exchange phase $R = (e^{i\pi})^{2s}$. The Gaussian's differential entropy $h = \frac{1}{2}\ln(2\pi e\sigma^2)$ carries both constants at once (verified: $\sigma = 1 \to 1.4189\ldots$, $\sigma = \frac{1}{2} \to 0.7258\ldots$, $\sigma = 2 \to 2.1121\ldots$), and the Jacobi theta constant $\vartheta_3(0; i) = 1.0864\ldots$ is the finite "total adelic information" of the Gaussian — the sum over the lattice of the fixed point (verified to 12 digits; the Jacobi identity $\theta(t) = t^{-1/2}\theta(1/t)$ holds to 10^{-14} at $t \in \{0.5, 1, 2, 3, 5\}$).

3. Statistics: The Missing Link

The word "statistics" is used in physics in two senses, and the adelic picture reveals they are the same thing at different places:

- **Quantum statistics** (spin-statistics): the exchange phase $R = e^{2\pi i s}$ of identical particles; bosons (s integer, $R = +1$) and fermions (s half-integer, $R = -1$) as the parity of $2s$ [2, 3, 13, 14, 17].
- **Probability statistics** (distribution theory): the occupation-number distributions of statistical mechanics — Bose–Einstein and Fermi–Dirac.

The link: **both are fixed points of maximum-entropy principles, distinguished by the exchange phase.** The Bose–Einstein distribution is the max-entropy distribution of occupation numbers with fixed mean energy; the Fermi–Dirac distribution is the same with the Pauli restriction $n \in \{0, 1\}$. What the exchange phase $R = e^{2\pi i s}$ does is *select which exponential modality applies* — the symmetric algebra (bosonic) or the exterior algebra (fermionic) — exactly the two modal exponentials whose braiding was formalized in the spin-statistics program [2, T1].

3.1 New result R1: p-adic maximum-entropy ≡ Bose–Einstein

Adelic Shannon Theory [1] defines p-adic entropy $H_p(X) = \sum_x p(x)\, v_p(x) = \mathbb{E}[v_p(X)]$ and identifies the maximum-entropy distribution as the geometric distribution on the valuation:

$$P(v_p(X) = k) = (1 - 1/p)\, p^{-k}, \qquad k = 0, 1, 2, \ldots$$

with $H_p^{\max} = 1/(p-1)$.

Now write the geometric distribution as a Bose–Einstein occupation distribution. The BE occupation-number distribution with fugacity $z = e^{-\beta(\varepsilon-\mu)}$ is

$$P(n) = (1-z)\, z^n, \qquad \langle n \rangle = \frac{z}{1-z}.$$

Setting $z = 1/p$ gives $P(k) = (1 - 1/p)\, p^{-k}$ — identical. The mean occupation is $\langle n \rangle = \frac{1/p}{1-1/p} = 1/(p-1)$ — identical to $H_p^{\max}$. **The p-adic maximum-entropy distribution is the Bose–Einstein distribution at fugacity $z = 1/p$, i.e. at inverse temperature $\beta_p = \ln p$.** The p-adic entropy is the mean occupation number of a bosonic mode at a temperature set by the prime itself. [ESTABLISHED — exact algebraic identity; verified numerically for $p = 2, 3, 5, 7$; means match $1/(p-1)$ to 10^{-12}.]

3.2 New result R2: squarefree restriction ≡ Fermi–Dirac

The fermionic counterpart of the valuation structure is the **squarefree restriction**: an integer is squarefree iff $v_p(x) \in \{0, 1\}$ for every prime p — each prime divides at most once. This is the p-adic analogue of the Pauli exclusion (occupation number per mode at most 1). Among squarefree integers, the probability that p divides x is

$$P(v_p = 1 \mid \text{squarefree}) = \frac{1}{p+1}$$

(a standard density fact: the squarefree density is $6/\pi^2$, and the p-divisibility condition removes the p^2 factor; the exact identity $(p-1)/(p^2-1) = 1/(p+1)$ is elementary). The Fermi–Dirac occupation number at fugacity $z = 1/p$ is

$$\langle n \rangle = \frac{1}{e^{\beta(\varepsilon-\mu)} + 1} = \frac{1}{z^{-1} + 1} = \frac{1}{p+1}.$$

Identical. The squarefree integers realize Fermi–Dirac statistics at the p-adic place, with the same inverse temperature $\beta_p = \ln p$. [ESTABLISHED — verified numerically over $N = 2{,}000{,}000$, 1,215,877 squarefree integers; the frequency of $p \mid n$ matches $1/(p+1)$ for $p = 2, 3, 5, 7$ with max deviation 8.53×10^{-6}.]

3.3 The dictionary

Archimedean (∞)	Non-archimedean (p)	Shared structure
Bose–Einstein $\langle n \rangle = 1/(e^{\beta(\varepsilon-\mu)} - 1)$	Geometric $P(k) = (1 - 1/p)p^{-k}$, $\langle v_p \rangle = 1/(p-1)$	Max-entropy with fixed mean; symmetric algebra
Fermi–Dirac $\langle n \rangle = 1/(e^{\beta(\varepsilon-\mu)} + 1)$	Squarefree $P(v_p = 1) = 1/(p+1)$	Pauli restriction; exterior algebra
$\beta = 1/k_B T$ (thermodynamic temperature)	$\beta_p = \ln p$ (prime logarithm)	Inverse temperature = fugacity logarithm
Boltzmann factor $e^{-\beta E}$	$p^{-k} = e^{-\beta_p k}$	Exponential of $-\beta\times$ "energy" (valuation)
Exchange phase $R = e^{2\pi i s} \in \{\pm 1\}$	Choice of valuation restriction (unrestricted / squarefree)	Statistics dichotomy

The p-adic "energy" is the valuation k; the p-adic "temperature" is the reciprocal prime logarithm; the p-adic "chemical potential" is implicitly absorbed into the fugacity normalization. The statistics dichotomy (boson/fermion) is mirrored exactly: unrestricted valuations = symmetric algebra = bosons; squarefree valuations = exterior algebra = fermions. The exchange phase $R = (e^{i\pi})^{2s}$ is the selector of the exponential modality at every place.

3.4 Tie-in to spin statistics and the tree program

The recent spin-statistics work established two things the adelic picture now absorbs:

1. $R = e^{2\pi i s} = (e^{i\pi})^{2s}$ **as a logical scalar**: the exchange phase is the $(2s)$-fold half-turn of the re-entrant mark [3]. In the present dictionary, the half-turn $e^{i\pi} = -1$ is the *fermion sign at the archimedean place*: $R = -1$ for $s = 1/2$, exactly as the squarefree restriction realizes the fermionic occupation channel at the p-adic places. The two statistics dichotomies are the same dichotomy in two places: the half-turn (archimedean exchange) and the valuation restriction (p-adic occupation) both select the exterior algebra over the symmetric algebra. [MAP — the arithmetic is exact; the physical identification is the model.]
2. **The p-adic anyon embedding**: the p-adic anyon program realizes braiding phases at roots of unity $\zeta_{2p^k} \mapsto e^{2\pi i/(2p^k)} = (e^{i\pi})^{1/p^k}$, i.e. rational spins $s = m/(2p^k)$ [3, §8]. Verified computationally: $\zeta_4 = i$, $\zeta_6 = e^{i\pi/3}$, $\zeta_8 = e^{i\pi/4}$ — the $(1/p^k)$-th roots of the half-turn. The rational-spin subsector of $R = (e^{i\pi})^{2s}$ is exactly the p-adic braiding phase lattice. [ESTABLISHED arithmetic; consistency of the two programs confirmed.]

The companion record One Table, Two Regimes [10] reads statistics as a tree-automorphism phase on the Bruhat–Tits tree, unifying the standard-model particle catalog with the condensed-matter excitation zoo. The valuation-restriction dictionary of this paper — unrestricted valuations = bosonic, squarefree = fermionic — is the companion occupation-statistics reading of the same non-archimedean statistics dichotomy; the two readings are consistent and mutually supporting.

4. Thermodynamics: The 2π and the Planckian Bound

4.1 The thermal Gaussian and LCI_opt

The Gaussian $e^{-\pi x^2}$ maximizes differential entropy at fixed variance — $h = \frac{1}{2}\ln(2\pi e\sigma^2)$ (verified: $\sigma = 1 \to 1.4189\ldots$, $\sigma = \frac{1}{2} \to 0.7258\ldots$, $\sigma = 2 \to 2.1121\ldots$). The constant $2\pi e$ appears as the entropy scale. The Quantum Architectonics program [4] derived the Lossless Complexity Index optimum $\mathrm{LCI}_{\mathrm{opt}} = \ln(2\pi) \approx 1.8379$ — the *natural log of the circle trace*. Both the Gaussian entropy and the architectural optimum carry the circle constant; in the re-entrant calculus, π is the trace of the identity on the circle type — the same π. [ESTABLISHED arithmetic; LCI_opt verified: $\ln(2\pi) = 1.837877\ldots$]

4.2 Planckian dissipation and the MSS bound

The Planckian dissipation bound (Maldacena–Shenker–Stanford [15]) states

$$\lambda_L \leq 2\pi\,\frac{k_B T}{\hbar},$$

with the Planckian scattering time $\tau_\hbar \approx \hbar/k_B T$. The numerical values (verified): at $T = 300$ K, $\lambda_{\max} = 2.468 \times 10^{14}\ \mathrm{s}^{-1}$, $\tau_\hbar = 2.546 \times 10^{-14}$ s; at $T = 77$ K, $\lambda_{\max} = 6.334 \times 10^{13}\ \mathrm{s}^{-1}$; at $T = 4$ K, $\lambda_{\max} = 3.29 \times 10^{12}\ \mathrm{s}^{-1}$.

The structural observation for the present synthesis: **the 2π in the MSS bound is the trace of the identity on the circle type.** The maximal Lyapunov exponent is $2\pi \times k_B T/\hbar$ — one circle-trace per thermal unit. The bound is the statement that dissipation cannot exceed one re-entrant turn per thermal time. The Planckian wall is the wall of the half-turn: $\tau_\hbar = \hbar/k_B T$ is the time in which the phase $e^{-iEt/\hbar}$ accumulates one radian of thermal rotation; the 2π bound is the full turn. [MAP — dimensional and structural identification; the identification of the MSS 2π with the circle trace is a reading, not a derivation.]

The companion program From Distinction to Dissipation [12] supplies the thermodynamics–statistics interface of the same program: second-law-gated braids and boundary costs, with a capacity ceiling $\lfloor \Delta S/(k_B \ln 2)\rfloor$ and a $2kT\ln 2$ inversion toll. The thermodynamic arm of this paper (β_p, Planckian 2π) extends that bridge to the p-adic place.

4.3 The p-adic temperature as a thermodynamic scale

The new results R1/R2 give the p-adic place a genuine thermodynamic reading: $\beta_p = \ln p$ is an inverse temperature. The p-adic entropy $H_p^{\max} = 1/(p-1) = \langle n\rangle_{\mathrm{BE}}(\beta_p)$ is a physical occupation number. The thermodynamic program of the Joules-per-Solution benchmark [5] — the energy cost of a correct quantum answer — gains a per-prime cost scale: the energy to resolve a p-adic digit is set by $k_B T_p$ with $T_p = 1/\ln p$ (in units where $\beta = 1/k_B T$). The prime $p = 2$ (binary digit) has $T_2 = 1/\ln 2 \approx 1.4427$ — the temperature of the smallest prime, hence the highest of all p-adic temperatures; higher primes are colder

($T_3 \approx 0.91, T_5 \approx 0.62$). [CONJECTURE — the thermodynamic interpretation of β_p as a physical temperature is proposed here for the first time; it is falsifiable via the conditions of Section 7.]

5. QND Measurement: The Equality Case

5.1 What QND is

A quantum non-demolition measurement measures an observable A without disturbing it: the measurement Hamiltonian commutes with A, so repeated measurements of A give the same result (the back-action is confined to the conjugate variable). QND is the workhorse of continuous quantum measurement and quantum metrology — cavity QED, gravitational-wave interferometry, and superconducting-qubit readout [18, 19, 20].

5.2 New result R3: QND = equality case of the DPI

The adelic data-processing inequality (Adelic Shannon Theory [1], Theorem 1′) states: for a p-adically contractive channel T, $H_p(T(X)) \leq H_p(X)$ — information cannot be created by p-adic processing. An ideal QND measurement of a p-adic-valued observable A is the **equality case**: the post-measurement distribution of A equals the pre-measurement distribution (the measurement does not disturb A), so

$$H_p(A \text{ after QND}) = H_p(A \text{ before}) \quad \text{exactly.}$$

Verified computationally: for the distribution $p = \{0.5, 0.25, 0.13, 0.06, 0.03, 0.02, 0.01\}$ over v_2, $H_p = 0.97$ before and after an ideal QND readout — invariant by construction, and this invariance is precisely the equality case of the inequality. The v1.1 verification adds a non-tautological contrast: over 200,000 seeded shots, the QND readout reproduces the exact pre-measurement valuation on every shot (readout fidelity 1.0), whereas a demolishing channel that redraws from the same marginal matches the true value only with probability $\sum_k p(k)^2 \approx 0.336$ — the demolition is information-lossy while QND is not. [ESTABLISHED — definitional; the content is the identification, which is exact, plus the contrast demonstration.]

The same holds at the archimedean place: QND preserves the full information vector $\mathbf{I}(X) = (I_\infty, I_2, I_3, \ldots)$ of the measured observable. **QND is the measurement family that saturates the data-processing inequality componentwise:** among measurement channels acting on A, the equality case (post-measurement distribution of A equal to its pre-measurement distribution) is precisely the non-demolition case. It is the information-conserving measurement: it extracts the readout without paying entropy in the measured channel. The category-theoretic record Valuation Without R [11] supplies a valuation-first foundation for finite measurement that is directly complementary to this reading.

5.3 The Born-rule boundary

The pre-registered falsification of deterministic measurement-triggered relaxation [6] showed that a deterministic map from a fixed initial state yields a degenerate outcome channel — measured probabilities take only the values 0 or 1 (max deviation 0.5, verified). Born statistics require one of three ingredients: an ensemble over initial states, stochasticity in the dynamics, or contextual hidden variables.

The QND connection: QND measurement is the *fourth* possibility that the falsification's three-ingredient taxonomy implicitly leaves open — not a relaxation at all, but an information-conserving readout of a pre-existing value. QND does not try to reproduce the Born rule from deterministic relaxation; it reads what is there and changes nothing. The Born statistics of a QND measurement are inherited from the preparation ensemble (ingredient 1) — which is why QND works: it adds zero measurement noise to the measured observable. The p-adic entropy conservation of R3 is the quantitative statement of this "zero measurement noise."

5.4 QND and the entropic number

The Measurement Stratigraphy [9] and Adelic Entropic Numbers [8] program defined the entropic number $(x, \mathbf{I}(X))$: a best estimate plus its full adelic information vector. The present synthesis gives QND measurement its natural data type: **a QND measurement of x returns the entropic number $(x, \mathbf{I}(X))$ with $\mathbf{I}(X)$ unchanged — the honest number, read without demolition.** The Gaussian $e^{-\pi x^2}$ is the universal entropic number (max-entropy at every place); QND is the measurement that preserves it. The two programs meet: entropic numbers are the data type of QND metrology. [MAP — structural identification; the experimental realization is future work.]

6. The Unification Map

Domain	Object	Fixed point	Scalar
Information theory	Gaussian $e^{-\pi x^2}$, Poisson summation	$\mathcal{F}[g] = g$ (Fourier self-duality)	$e^{-\pi}$, $\vartheta_3(0; i) = 1.0864\ldots$
Statistics (quantum)	Exchange phase $R = e^{2\pi i s}$	$(e^{i\pi})^{2s}$ — half-turn power	$e^{i\pi} = -1$
Statistics (thermal, ∞)	Bose–Einstein / Fermi–Dirac	Max-entropy with fixed mean	$\beta = 1/k_B T$
Statistics (thermal, p)	Geometric / squarefree valuation	Max-entropy with fixed mean valuation	$\beta_p = \ln p$
Thermodynamics	Boltzmann factor $e^{-\beta E}$, Planckian bound	Max-entropy variational	2π (circle trace), $\lambda_L \leq 2\pi k_B T/\hbar$
QND measurement	Information-conserving channel	Equality case of DPI (H_p preserved)	$\mathbf{I}(X)$ conserved

Every row is a fixed point of a self-referential operation; every scalar is a member of the re-entrant family (e as fixed point of D, π as circle trace, $R = (e^{i\pi})^{2s}$ as monodromy power); the half-turn $e^{i\pi} = -1$ is the shared generator (statistics at the archimedean place, fermion sign, the exchange dichotomy); the p-adic places realize the same structure with $\beta_p = \ln p$.

The relationship, in one sentence: information theory, statistics, thermodynamics, and QND measurement are the four faces of the self-referential scalar family of the re-entrant mark — e (local self-reference: $Df = f$), π (global self-reference: the circle's trace), and their combination $e^{i\pi} = -1$ (the half-turn), whose $(2s)$-fold power is the exchange phase that selects the statistics, whose thermal realization is the Boltzmann/max-entropy family with the Planckian bound as its dissipation ceiling, and whose measurement theory is the QND channel that conserves the information vector exactly.

7. Falsifiability and Status Ladder

Status ladder

Claim	Status
R1: p-adic max-entropy ≡ Bose–Einstein at $z = 1/p$; $H_p^{\max} = \langle n \rangle = 1/(p-1)$	[ESTABLISHED — exact identity, verified $p = 2, 3, 5, 7$]
R2: squarefree restriction ≡ Fermi–Dirac; $P(v_p = 1) = 1/(p+1)$	[ESTABLISHED — verified over 1.2M squarefree integers]
$\beta_p = \ln p$ is an inverse temperature at the p-adic place	[MAP — exact algebra; physical temperature reading]
R3: ideal QND saturates the adelic DPI componentwise	[ESTABLISHED — definitional identification]
MSS 2π is the circle trace; Planckian wall = half-turn wall	[MAP — dimensional/structural reading]
$T_p = 1/\ln p$ is a physical energy scale (energy per prime digit)	[CONJECTURE — new, falsifiable below]
The four domains are one fixed-point family	[CONJECTURE — the unification thesis]

Falsifiability conditions

- **F1 (thermal p-adic scale).** If a physical process is exhibited whose energy cost per resolved p-adic digit scales other than $\sim k_B T_p = k_B / \ln p$ (or with a different prime ordering than $T_2 > T_3 > T_5 > \ldots$), the thermodynamic reading of β_p is disconfirmed. The ordering $T_2 \approx 1.44 > T_3 \approx 0.91 > T_5 \approx 0.62$ is the sharpest prediction.
- **F2 (QND conservation).** If a measurement is exhibited that is QND for a p-adic-valued observable (commutes with it, repeatable) yet changes H_p of that observable, R3 is disconfirmed. (This is definitionally impossible for ideal QND; the test applies to any claimed "effective QND" realization.)

- **F3 (statistics–valuation dictionary).** If a physical system with p-adic-valued observables exhibits occupation statistics at the p-adic place that are neither geometric (bosonic) nor squarefree-Bernoulli (fermionic) — e.g. a q-analogue with $q \neq 1/p$ — the dictionary of Section 3.3 is disconfirmed for that system. (The dictionary remains a classification claim: it does not predict which systems realize which statistics.)
- **F4 (unification thesis).** If a member of the scalar family (e, π, R) is shown to arise in one of the four domains from a *non*-self-referential origin that is incompatible with the re-entrant generation, the unification thesis is weakened to a correspondence. The thesis is strengthened by each independent derivation of the same constant from distinction.

8. What a Practitioner Can Do With This

1. **QND metrology with p-adic observables.** Any quantum sensor whose observable has number-theoretic structure (photon numbers, flux quanta, harmonic-oscillator levels) can carry a per-prime uncertainty budget: H_p is conserved by QND readout, so the p-adic digits of the measurement are the noise-free channel. The information vector $\mathbf{I}(X) = (I_\infty, I_2, I_3, \ldots)$ is the complete metrological error budget — a concrete specification for sensor calibration.
2. **p-adic noise models in quantum engineering.** The AUM channel with $\beta_p = \ln p$ gives a closed-form capacity $C_p = \log_p(1 + \mathrm{SNR}_p)$ for noise whose valuation structure is prime-specific; the doubling $C_2 = 2C_\infty$ (verified for all SNR) is a design rule for binary-resolved systems.
3. **Energy benchmarking per prime digit.** The Joules-per-Solution metric [5] gains a place-wise decomposition: the energy to resolve a p-adic digit is set by $k_B T_p = k_B / \ln p$; a benchmark that reports energy per prime digit is directly comparable across architectures.
4. **Planckian design rule.** The MSS bound $\lambda_L \leq 2\pi k_B T/\hbar$ with the circle-trace reading gives architects of strongly correlated systems a dimensionless target: $\mathrm{LCI} = \ln(2\pi)$ is the optimal structural complexity; the 2π is not a convention but the trace of the identity on the circle type — the same constant in the re-entrant calculus and in the dissipative bound.
5. **Measurement-channel accounting.** The QND equality case gives a practical audit rule for measurement chains: any measurement that reduces H_p of the measured observable is non-QND (it has demolition back-action); the information loss is exactly $H_p(\text{before}) - H_p(\text{after})$, computable in situ.

9. Relation to the Prior Work

- Adelic Shannon Theory [1]: supplied H_p, the AUM channel, the product-formula coding theorem, the Gaussian as universal max-entropy function. This paper adds the statistical-mechanical reading (R1, R2), the QND equality case (R3), and the thermodynamic arm (β_p, T_p).
- The Exchange Phase as a Logical Scalar [3] and The Boson/Fermion Distinction [2]: supplied $R = (e^{i\pi})^{2s}$, the two-modal-exponential construction, the p-adic anyon embedding. This paper ties those to the occupation-number distributions (the exponentials *are* the BE/FD distributions at the p-adic place) and to the Planckian bound.

- One Table, Two Regimes [10]: reads statistics as a tree-automorphism phase on the Bruhat–Tits tree, unifying the standard-model particle catalog with the condensed-matter excitation zoo. This paper's valuation-restriction dictionary is the companion occupation-statistics reading of the same non-archimedean statistics dichotomy; the two records are consistent and mutually supporting.
- Valuation Without R [11]: supplies a category-theoretic foundation for finite measurement without the real numbers. This paper's R3 is the measurement-theoretic statement in the same valuation-first direction; the two records should be cited together.
- From Distinction to Dissipation [12]: second-law-gated braids and boundary costs at the thermodynamics–statistics interface of the same program. This paper's thermodynamic arm (β_p, Planckian 2π) extends that bridge to the p-adic place.
- The Calculus of Re-Entrant Distinctions [7]: supplied e and π as logical scalars of the mark. This paper adds the third and fourth faces (Fourier fixed point; max-entropy fixed point) and the thermodynamic 2π.
- Measurement Stratigraphy [9] and Adelic Entropic Numbers [8]: supplied entropic numbers and the observer-relativity reading. This paper identifies QND as the measurement that realizes entropic numbers without demolition.
- Quantum Architectonics / Planckian Dissipation [4]: supplied $\mathbf{LCI}_{\mathrm{opt}} = \ln(2\pi)$ and the Signal-Worker ontology. This paper connects $\mathbf{LCI}_{\mathrm{opt}}$ to the circle trace and the MSS 2π to the half-turn.
- A Pre-Registered Falsification of Deterministic Measurement-Triggered Relaxation [6]: supplied the three-ingredient taxonomy of Born statistics. This paper adds QND as the information-conserving fourth path and the H_p-conservation criterion.
- External anchors: Shannon [16], Pauli [13], Leinaas–Myrheim [14], Wilczek [17], Maldacena–Shenker–Stanford [15]; QND characterization and certification [18, 19, 20].

10. Conclusion

Information theory, statistics, thermodynamics, and QND quantum measurement are organized by one self-referential scalar family — e, π, and $R = (e^{i\pi})^{2s}$ — generated by the re-entrant mark and realized at every place of the rationals. The p-adic maximum-entropy distribution is a Bose–Einstein distribution at $\beta_p = \ln p$; the squarefree integers are its Fermi–Dirac counterpart; the exchange phase selects the statistics; the Planckian bound is the circle trace in thermal units; and QND measurement is the equality case of the data-processing inequality — the channel that conserves the information vector exactly. The premises end where the identification of a *physical* temperature at the p-adic place begins: the algebra is exact, the physics is proposed, and the falsification conditions are written.

Declarations

- **Funding:** This research received no external funding.
- **Conflicts of interest:** The author declares no conflicts of interest.
- **Verification:** All numerical claims verified by `artifacts/verification/adelic-stats-verification.py` (deposited source; stdlib-only CPython 3; deterministic — seeded Monte Carlo (seed 20260820,

200,000 shots) only for the QND readout-fidelity contrast; squarefree sieve N = 2×10^6; output `adelic-stats-verification-2026-08-20.json`, 38 checks, all pass). Reproducibility: `python artifacts/verification/adelic-stats-verification.py` re-generates the JSON byte-identically (run log: `run-2026-08-20.txt`).
- **AI assistance:** AI-assisted drafting and verification orchestration; all computational results produced by the deposited executed code.
- **Preprint policy:** Self-archived working paper; not submitted for journal peer review (standing venue policy: Zenodo).

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